



# Single aperiodically poled lithium niobate chip for simultaneous electro-optic spectral and Q switchings in a cross-polarized, closely-lying dual-wavelength Nd:YLF laser

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**Abstract:** We report on a fast switchable light source in a Nd:YLF laser oscillating at two cross-polarized, closely lying emission lines ( $\pi$ -polarized 1047-nm and  $\sigma$ -polarized 1053-nm lines), achieved using an intracavity aperiodically poled lithium niobate (APPLN) electro-optic (EO) photonic chip. The APPLN, whose domain structure is built with a simulated annealing optimization algorithm, functions as a wavelength- and voltage-dependent polarization mode converter to manipulate the polarization-dependent laser gain of the dual emission lines in Nd:YLF. When the APPLN operates at characteristic voltages,  $V_{HQ} = 0, 572, \text{ and } 723 \text{ V}$ , dual cross-polarized 1047- and 1053-nm lines, single  $\pi$ -polarized 1047-nm line, and single  $\sigma$ -polarized 1053-nm line will be gained (i.e.,  $V_{HQ}$  is a voltage turning the cavity into a high-Q state for a specific laser line(s)) in the laser, respectively, while the gain of both lines will be suppressed at  $V_{LQ} = 157 \text{ V}$ . Fast EO switching among the three different laser spectra (of different polarization states) can thus be done simply by tuning the APPLN operating voltage. Even interestingly, when modulating the APPLN with a voltage pulse train switching between a  $V_{HQ}$  and  $V_{LQ}$ , we observed a few tens of ns, high peak-power ( $\sim 20 \text{ kW}$ ) laser pulses from the system for the three output spectra (along with their relatively weak non-phase-matched second harmonic and sum frequency signals generated through the APPLN), again switchable with respect to their characteristic voltages  $V_{HQ}$ . This system potentially features multiplexability in time, wavelength, and polarization.

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## 1. Introduction

Coherent light sources with a multiline spectrum have been of big interest to many applications including optical communications, spectroscopy, LiDAR, chemical sensing, and biomedical imaging [1–3], thanks to the increase of the processing speed and multiplexity of the systems without the need for multiple sources (of different wavelengths) or resorting to the fast tuning of the source spectrum. The ability to fast switch among the spectral lines of a multi-wavelength source is also desirable to further extend its capacity to benefit the development of, e.g., advanced LiDAR and optical network systems [4,5]. Switchable multi-wavelength sources have been demonstrated in lasers with a relatively broad gain bandwidth, such as erbium-doped fiber lasers and diode lasers, via the modulation of the built (Bragg) wavelength-selective conditions or the tuning of the engineered laser gain/resonant configurations, respectively [6,7]. However, the former switching mechanism has been based on a slow mechanical control, while the latter one could be fast (ns regime) but with a relatively limited tuning range (so that an integration of various gain sections is required to extend the working spectrum). Besides, the demonstrated switchable diode lasers are usually characterized by a low (mW level) output power, reducing their applicability (e.g.,

in remote sensing). High-peak-power pulsed sources have been widely implemented in fiber lasers as well as solid-state lasers. In contrast, solid-state lasers can feature better compatibility to integrate with various optical elements such as prisms, gratings, filters, or etalons [8] to accomplish a desired tunability (within the gain bandwidth of an emission line or among the different transient lines), though still relying on a slow tuning mechanism. Besides, the spectral tailorability of those dispersive or interference elements is usually limited, lowering their ability to select/switch a group of (two or more) different transition lines at the same time in a multiline laser system and, therefore, reducing their versatility. Even more challenging issues could arise for those mechanisms when a desired processing (switching/selection/modulation/tuning) of a light is polarization-dependent.

The combination of electro-optic (EO) effects and a solid-state laser technique could be a viable scheme to address the above issues. EO effects are usually acquired based on a quadratic ( $\chi^{(2)}$ ) nonlinear optical material and could be the most prominent mechanism used to realize high-speed optical signal processing devices [9]. EO devices have also been widely employed in solid-state laser systems to function as, e.g., laser Q-switches to produce energetic fast (on the order of ns) light pulses [10–12]. Lithium niobate (LN) has been one of the most popular EO substrates thanks to its superior nonlinearity [13]. Besides a common practice based on refractive-index change of LN (i.e., Pockels cells), we have developed another class of EO Q-switches with access to the off-diagonal Pockels tensor component,  $r_{42}$  ( $=r_{51}$ ), of a periodically poled LN (PPLN) having a domain length equal to  $\lambda/2(n_o - n_e)$ , where  $\lambda$  is the wavelength and  $n_o$  and  $n_e$  are refractive indices of the ordinary (*o*) and extraordinary (*e*) polarization modes excited in the crystal [14]. Such a PPLN device is then equivalently composed of a stack of half-wave plates whose optic axes alternatively rotate with respect to the crystallographic *z* axis with an angle proportional to the strength of the applied electric field  $E_y$  along the crystallographic *y* axis, functioning as an EO polarization rotator or, in particular, a polarization mode converter (PMC, converting between *o* and *e* modes) when a specific voltage, called half-wave voltage,  $V_\pi = \lambda d / (2\gamma n_o^{3/2} n_e^{3/2} r_{42} L)$  is used to build  $E_y$ , where  $d$  is the spacing between the electrodes formed on the two *y* surfaces of the PPLN,  $\gamma$  is the nonlinear reduction factor arising from the quasi-phase-matching (QPM) domain structure of the device ( $\gamma = 2/\pi$  for a PPLN), and  $L$  is the device length. The laser cavity Q factor can thus be fast modulated when such an EO device operates in a laser oscillator with a polarization-dependent/sensitive gain [15,16].

Besides playing as a fast polarization controller (controllability in “polarization” domain and fast modulability in “time” domain), such an EO PMC can work as a narrowband ( $\sim 1$ – $2$  nm·cm at the near-IR band [14]) spectral filter (selectivity in “spectral” domain) thanks to its notch-like conversion spectrum, in which we have demonstrated several intriguing EO spectral tailorable light sources [17,18]. More recently, we have utilized such a unique spectral selectivity featured in the device to realize pulsed, EO switchable coherent light sources in a multiline optical parametric oscillator, as well as in a dual-wavelength (1064 and 1342-nm) Nd:YVO<sub>4</sub> laser [19,20]. Nd<sup>3+</sup> lasers are the most widely known efficient solid-state lasers, having the strongest emission lines at  $\sim 1$   $\mu$ m, and have found various applications in fields such as medicine, materials processing, and the military. Among the Nd<sup>3+</sup> lasers, Nd:YLF is of particular interest because of its unique optically anisotropic (or birefringent) properties that enable the generation of cross-polarized dual-wavelength spectrum in an emission band, such as the  $^4F_{3/2} \rightarrow ^4I_{11/2}$  transition where a  $\pi$ -polarized 1047-nm line and a  $\sigma$ -polarized 1053-nm line with similar stimulated emission cross sections ( $1.8 \times 10^{-19}$  and  $1.2 \times 10^{-19}$  cm<sup>2</sup>, respectively) have been observed [21]. Besides, Nd:YLF also benefits from its relatively long fluorescence lifetimes to produce energetic Q-switched laser pulses [22].

Therefore, in this work, we further extend our study of the EO PMC in a QPM LN device to achieve simultaneously the EO spectral and Q switchings as well as nonlinear wavelength conversion in a dual-wavelength (1047 and 1053 nm) Nd:YLF laser. To the best of our knowledge,

this could be the first demonstration of an EO switchable laser working between two cross-polarized and closely lying (within a few nm) spectral lines. Besides the wavelength and time, such a light source is endowed with another degree of freedom in polarization to expand its multiplexing, which is of great value for potential applications such as optical coherence tomography, holography, THz generation, and precision sensing [23–25].

## 2. Multiline EO PMC device design, simulation, and fabrication

The unique idea developed in this study to achieve the fast switching between the two orthogonally-polarized radiation lines (at 1047 and 1053 nm) of a Nd:YLF laser is to modulate the polarization-dependent gain of the laser via the EO control of an intracavity wavelength-dependent QPM PMC which is the core device created in this work. Specifically, the QPM PMC device is designed to be able to operate at three characteristic voltages,  $V_{\lambda_\pi}$ ,  $V_{\lambda_\sigma}$ , and  $V_{LQ}$ , where at  $V_{\lambda_\pi}$  the device will highly transmit the  $\pi$ -polarized 1047-nm line (i.e., its polarization state and therefore the laser-gain access remain) but convert the polarization state of the 1053-nm line from  $\sigma$  to  $\pi$  after a cavity round-trip transit (and therefore loss of the laser gain), while at  $V_{\lambda_\sigma}$  the device will operate on the polarization states of the two lines on the other way round, i.e., the  $\sigma$ -polarized 1053-nm line will remain its polarization state and the access of the laser gain but the 1047-nm line will lose its laser gain as its polarization is converted from  $\pi$  to  $\sigma$  state through the device in the laser cavity. Therefore, a Nd:YLF laser can then be switchable among three emission spectra containing cross-polarized 1047- and 1053-nm dual lines (the initial/default spectrum),  $\pi$ -polarized 1047-nm single line, and  $\sigma$ -polarized 1053-nm single line when such a unique intracavity EO QPM PMC device operates at  $V_{EO} = 0$ ,  $V_{\lambda_\pi}$ , and  $V_{\lambda_\sigma}$ , respectively. These three emission conditions correspond to the high Q states of the Nd:YLF laser for the oscillation of the dual 1047- and 1053-nm lasers, single 1047-nm laser, and single 1053-nm laser, respectively.

More interestingly, in our design, both emission lines of the Nd:YLF laser can be suppressed when the EO QPM PMC device operates at  $V_{EO} = V_{LQ}$ , which corresponds to a common low Q state for the dual laser lines. This enables the generation of so-called giant pulses from the Nd:YLF laser through the Q-switching mechanism, achieved here using the EO control of the QPM PMC device. Specifically, the generation of Nd:YLF laser pulses of dual wavelengths at 1047 and 1053 nm, a single wavelength at 1047 nm, and a single wavelength at 1053 nm can be realized when the intracavity EO QPM PMC device is driven by a fast ( $\sim$ ns) voltage pulser operating between  $V_{EO} = V_{LQ}$ , leading to the low Q state of the laser, and  $V_{EO} = 0$ ,  $V_{\lambda_\pi}$ , and  $V_{\lambda_\sigma}$ , leading to the high Q states for the lasing of the three specific spectra as discussed above, respectively. We summarize in Tab. 1 the operation conditions of the EO QPM PMC device in a Nd:YLF laser to achieve a fast cw and pulsed switchable coherent light source based on its cross-polarized, closely-lying emission lines, which also reveals the design principle of our EO QPM PMC device.

According to Tab. 1, the EO QPM PMC device designed to implement a switchable Nd:YLF laser should meet all the following operating conditions, referred to hereinafter as the “polarization-dependent gain controllable” (PDGC) conditions, in a single-chip operation, comprising “gain suppressed” or “low-Q” state conditions: the conversion efficiency of the device should be high enough at  $V_{EO} = V_{\lambda_\sigma}$  and  $V_{LQ}$  for the 1047-nm line and at  $V_{EO} = V_{\lambda_\pi}$  and  $V_{LQ}$  for the 1053-nm line to convert their polarization states to those that will experience a major gain loss, and “gained” or “high-Q” state conditions: the polarization states should be maintained without conversion in the device when operating at  $V_{EO} = 0$  V and  $V_{\lambda_\pi}$  for the 1047-nm line and at  $V_{EO} = 0$  V and  $V_{\lambda_\sigma}$  for the 1053-nm line to remain their access of the laser gain. Since the conversion efficiency of an EO QPM PMC under the interaction with a wave at a phase-matched wavelength  $\lambda$  is a function of the driving voltage  $V_{EO}$ , given by

$$\eta_{EOP} = \sin^2(|\kappa|L), \quad (1)$$

**Table 1. Operation conditions/schemes of the EO QPM PMC device in a cross-polarized, dual-wavelength Nd:YLF laser to achieve a fast cw and pulsed switchable coherent light source**

| Operating voltage (condition)    | CW switching        |                        |                           | Q switching         |                        |                           | Emission spectrum (schematic) | Voltage tuning curve (schematic) |
|----------------------------------|---------------------|------------------------|---------------------------|---------------------|------------------------|---------------------------|-------------------------------|----------------------------------|
| $V_{EO}$                         | Laser line          |                        | Possible operating scheme | Laser line          |                        | Possible operating scheme |                               |                                  |
|                                  | $\pi$ -pol. 1047 nm | $\sigma$ -pol. 1053 nm |                           | $\pi$ -pol. 1047 nm | $\sigma$ -pol. 1053 nm |                           |                               |                                  |
| $V_0 = 0$ V (cond. I)            | ✓                   | ✓                      |                           | HQ                  | HQ                     |                           |                               |                                  |
| $V_{\lambda_\pi}$ (cond. II)     | ✓                   | ×                      |                           | HQ                  | LQ                     |                           |                               |                                  |
| $V_{\lambda_\sigma}$ (cond. III) | ×                   | ✓                      |                           | LQ                  | HQ                     |                           |                               |                                  |
| $V_{LQ}$ (cond. IV)              | ×                   | ×                      |                           | LQ                  | LQ                     | n.a.                      |                               |                                  |

✓: gained, ×: gain suppressed, HQ: high-Q state, LQ: low-Q state

where  $\kappa = G(\pi/\lambda)(n_o n_e)^{3/2} r_{42} V_{EO} / d$  with  $G$  being the nonlinear reduction factor associated with the Fourier transform of the QPM domain polarity structure [14], it is thus possible to tune the behavior, or the periodicity of the sine-squared varied, voltage-dependent efficiency of the device through the manipulation of an appropriate  $G$ , which is determined by the domain structure of the device. For example, an EO QPM PMC device, characterized by two different voltage tuning curves for two cross-polarized  $\lambda_\pi$  and  $\lambda_\sigma$  waves as schematically shown in the inset (referred to as “Fig. T1”) in the rightmost column of Tab. 1, if created, can possibly meet the PDGC conds. I-IV (see Tab. 1) at  $V_{EO} = 0$  (where both curves are at a peak position),  $V_{\lambda_\pi}$  (where the  $\pi$ -polarized curve is at a peak position, while the  $\sigma$ -polarized curve is at a trough position),  $V_{\lambda_\sigma}$  (where the  $\sigma$ -polarized curve is at a peak position, while the  $\pi$ -polarized curve is at a trough position), and  $V_{LQ}$  (where both curves are at a trough position), respectively.

To achieve fast switching among the voltage-dependent spectra in the laser as proposed in Tab. 1, we calculate and pursue an optimum domain structure for the EO QPM PMC device to function in all the operating scenarios designed in Tab. 1 with as low as possible operating voltages,  $V_{\lambda_\pi}$ ,  $V_{\lambda_\sigma}$ , and  $V_{LQ}$ , by a computer code developed based on the aperiodic optical superlattice (AOS) technique [26] and optimized by the simulated annealing (SA) method [27]. This is to produce two reciprocal vectors (associated with nonlinear reduction factors,  $G_{\lambda_\pi}$  and  $G_{\lambda_\sigma}$ ) here in a domain-engineered LN chip to quasi-phase-match the EO PMC processes for the dual emission lines of a Nd:YLF laser, respectively, to satisfy all the multiple EO PDGC conditions as discussed above. This is equivalent to constructing an AOS LN (i.e., an aperiodically poled LN (APPLN)) to perform as a dual-wavelength EO PMC device featuring two operating (voltage) tuning curves of adequate periodicities (determined by  $G_{\lambda_\pi}$  and  $G_{\lambda_\sigma}$  according to Eq. (1)) for the two wavelengths (1047 and 1053 nm), respectively, as schematically illustrated in Fig. T1.

To design such a device in an APPLN with the SA algorithm, we create a LN chip scheme, which is composed of an array of  $N$  unit domain blocks, to be the basis of the domain structuring

and optimization. The domain polarities of the  $N$  blocks are determined by the SA method with steps emulating a thermal annealing process that has been used to refine an optimized lattice structure of a solid-state material. The process starts with a high “temperature” (randomness) to broaden the solution space for search, and then properly controls the “cooling” rate to reduce the probability of accepting worse solutions and converges the search on better ones.

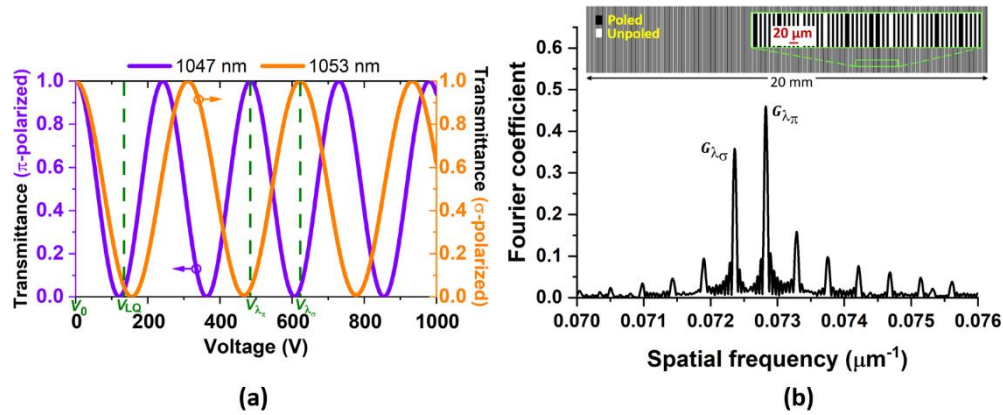
In our SA algorithm, we begin the annealing process with an initial solution, where the polarity of all the  $N$  unit domain blocks is set to be positive. Then the initial solution is perturbed by sequentially flipping the polarity orientation of each unit block in the LN chip. For every flip, the conversion efficiency of a new solution (a new domain structure) is calculated and is evaluated using an objective function ( $OF$ ). If it is evaluated to be a better solution (i.e., conversion efficiency is improved), it is accepted as the new solution. If it turns out to be a worse solution, it may still be accepted with a certain probability set in the algorithm. Such a probability mechanism provides an opportunity to escape from local optima, thereby increasing the likelihood of finding the global optimum. An iteration is complete when the polarities of all domain blocks are determined through the above process, which then serves as the new initial solution for the next iteration. After each iteration, the temperature is decreased according to the set cooling schedule to facilitate progress to a convergent solution. The calculation continues until a set “low temperature” or a default number of iterations is reached, which ends the SA algorithm and yields an optimized APPLN domain structure. The  $OF$  used in this study is formulated as

$$OF(V_{LQ}) = \left( \sum_{j=1}^M w_{\eta}(\lambda_j) \left| \eta_{EOP,i}(\lambda_j, V_{LQ,i}) - \eta_{EOP,c}(\lambda_j, V_{LQ,i}) \right| \right) + \gamma_{\eta} \left( \max[\eta_{EOP,c}(\lambda_1), \dots, \eta_{EOP,c}(\lambda_M)] - \min[\eta_{EOP,c}(\lambda_1), \dots, \eta_{EOP,c}(\lambda_M)] \right), \quad (2)$$

where  $M = 2$  is used here for considering the operation of the two Nd:YLF laser lines ( $\lambda_1 = 1047$  nm and  $\lambda_2 = 1053$  nm),  $\eta_{EOP,i}(\lambda_j, V_{LQ,i})$  and  $\eta_{EOP,c}(\lambda_j, V_{LQ,i})$  are the target and calculated roundtrip conversion efficiency for the building APPLN EO PMC device when operating at  $\lambda_j$  and  $V_{LQ,i}$ ; here  $\eta_{EOP,c}$ , instead of being obtained directly from the form shown in Eq. (1), is calculated based on the solving of the coupled-mode theory [28] to allow for the iterative search of an optimized polarity structure for the  $N$  domain blocks one by one using the SA algorithm,  $w_{\eta}(\lambda_i)$  is the efficiency weighting factor for the  $i$ -th conversion process, and  $\gamma_{\eta}$  is the weighting factor for equalizing the efficiency of the multiple conversion processes, and the operators  $\max[\dots]$  and  $\min[\dots]$  output the maximum and minimum values enclosed in the bracket. With  $V_{LQ,i}$  as an input parameter, the  $OF$  in Eq. (2) aims to guide the algorithm to find the common highest  $\eta_{EOP,c}$  for the two cross-polarized laser lines at a proper but lowest possible voltage, which yields  $V_{LQ}$ , determined directly by examining the voltage tuning curves of the APPLN EO PMC with domain structures calculated in a range of  $V_{LQ,i}$  (0–200 V) to find one that can work to best meet all the PDGC conditions with operating voltages,  $V_{\lambda_{\pi}}$  and  $V_{\lambda_{\sigma}}$ , being as low as possible as well. Operating at a common low Q state for the two cross-polarized laser lines is one attempt to equalize the cavity conditions, thereby achieving a more gain-balanced dual-wavelength Q-switched laser. In the algorithm, we have set an upper limit of 800 V (applied across a 1-mm electrode gap) for the calculation of those voltages, considering the experimental availability in our lab. In this study, a device length of 2 cm is used, and the thickness ( $\Delta x$ ) of each unit domain block is set to 5  $\mu\text{m}$  to accommodate our technical capability in a short-period domain poling in bulk LN, which gives  $N = 4000$ , though in general, a finer  $\Delta x$  or equivalently, a larger  $N$  should facilitate the algorithm to compose a more optimized domain structure, as it provides more building blocks to allow for creating more diverse domain structures for evaluation.

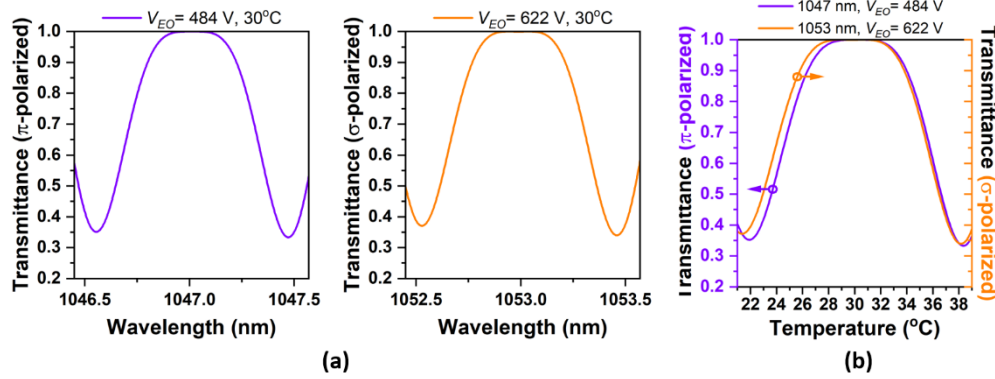


Figure 1(a) shows the characteristic voltage tuning curves of the resultant APPLN calculated from the developed SA algorithm for the roundtrip transmittance ( $\equiv 1 - \eta_{EOP}$ ) of a  $\pi$ -polarized 1047-nm (purple curve) and a  $\sigma$ -polarized 1053-nm (orange curve) waves, respectively, which are equivalent to the gain (or relatively, loss) modulation curves when it operates in the Nd:YLF laser having a polarization-dependent gain. Since the two cross-polarized lines won't interact in the APPLN, the two voltage tuning curves shown in Fig. 1 can be calculated individually. This optimal result is found when  $w_\eta(\lambda_i) = 1$ ,  $\eta_{EOP,i} = 1$ , and  $\gamma_\eta = 0.3$  are used in the algorithm. By referring to Fig. T1 in Tab. 1, we can readily identify from Fig. 1(a) that  $V_{LQ} \sim 133$  V,  $V_{\lambda_\pi} \sim 484$  V, and  $V_{\lambda_\sigma} \sim 622$  V with a peak transmittance  $\sim 1$  for all gained/high-Q states and an ultra-low transmittance  $< 0.038$  for all gain suppressed/low-Q states, as indicated by green dashed lines. Though a certain loss of gain might already cease the laser, we still pursue in low-Q states a highly gain-suppressed ratio (i.e.,  $> 0.962$ ) in the design to show an ultimate capability of our device. We Fourier analyze the domain structure of the calculated APPLN (as schematically represented in the inset of Fig. 1(b) by a bar-code-like structure) and find that two major spectral peaks are resolved at  $0.0728$  and  $0.0723 \mu\text{m}^{-1}$ , exactly corresponding to the spatial frequencies associated with the reciprocal vectors required for phase-matching the EO PMC processes at 1047 and 1053 nm, respectively, at  $30^\circ\text{C}$ , as shown in Fig. 1(b). The ratio of  $G_{\lambda_\sigma}$  to  $G_{\lambda_\pi}$  is  $\sim 0.776$ , agreeing with the periodicity ratio of the two tuning curves shown in Fig. 1(a).



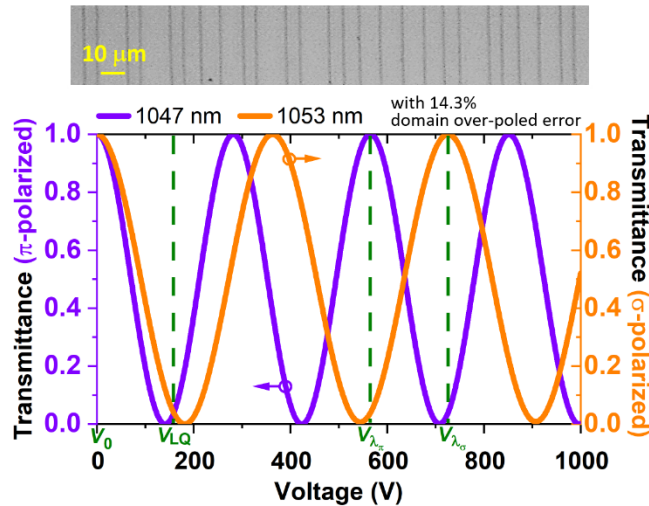
**Fig. 1.** (a) Calculated voltage tuning curves of the built APPLN EO PMC for the roundtrip transmittance ( $\equiv 1 - \eta_{EOP}$ ) of a  $\pi$ -polarized 1047-nm (purple curve) and a  $\sigma$ -polarized 1053-nm (orange curve) waves. (b) Fourier spectrum of the calculated APPLN domain structure (as displayed in the inset), where the two major spectral peaks correspond to the spatial frequencies required for phase-matching the dual EO PMC processes at 1047 and 1053 nm, respectively.

It is also interesting to investigate the operating bandwidths of such a multi-line EO polarization-controlling device. Figures 2(a) and 2(b) show the calculated tuning results for the EO APPLN device with respect to the operating wavelength and temperature, respectively, for the switching voltages at  $V_{\lambda_\pi} = 484$  V and  $V_{\lambda_\sigma} = 622$  V, as identified from Fig. 1(a). The results reveal a high-Q state (corresponding to a transmittance  $\sim 1$ ) can still be maintained for the two laser lines in a spectral bandwidth of  $\sim 0.2$  nm and a temperature bandwidth of  $\sim 2.4^\circ\text{C}$  when performing the spectrum switching with the device in a Nd:YLF laser at  $V_{\lambda_\pi}$  and  $V_{\lambda_\sigma}$ , respectively. Besides, it can be found the EO APPLN device exhibits a conversion bandwidth (full width at half maximum (FWHM)) of  $\sim 0.72$  nm, implying this unique device could additionally provide a spectral narrowing mechanism in the Nd:YLF laser featuring a broader fluorescence spectrum of  $\sim 1.5$ -nm bandwidth (FWHM) [29].



**Fig. 2.** (a) Wavelength and (b) Temperature tunings of the transmittance of the APPLN EO PMC device at characteristic switching voltages of  $V_{\lambda\pi} = 484$  V and  $V_{\lambda\sigma} = 622$  V.

We then fabricated the EO APPLN device by implementing the calculated aperiodic domain structure in a 20-mm-long, 1-mm-wide, and 0.5-mm-thick  $z$ -cut LN chip using the photolithography and electric-field poling techniques [30]. The domain structure of the fabricated APPLN chip was revealed by the HF etching method applied to the  $+z$  surface of the crystal (etching rates for domains of opposite polarity are different, thereby contrasting the poled and unpoled domain structures on the crystal surface). The inset of Fig. 3 shows a representative microscopic image of a portion of the etched domain structure, revealing a clear-cut domain pattern but exhibiting a nonuniformly distributed domain over-poled error. It is not rare to observe such a domain poling error originating from the domain-wall expansion dynamics during the electric-field poling process [31], resulting in the decrease of the nonlinear reduction factor  $G$  and therefore the increase of the periods of the tuning curves. Figure 3 shows the simulated voltage tuning curves of the APPLN EO PMC when its original (i.e., as calculated) domain structure is modified to have a (uniform) 14.3% over-poled error. Here, an equivalent 14.3% is adopted as it is found to

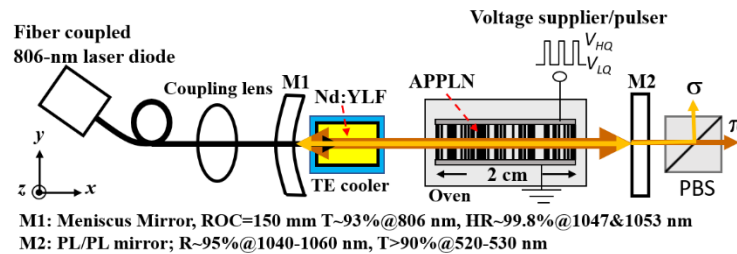


**Fig. 3.** Simulated voltage tuning curves of the APPLN EO PMC with a 14.3% domain over-poled error. The inset shows a microscopic image of a portion of the etched domain structure.

best fit the measurement results (see below) and can be regarded as the average value of the poled APPLN domain error. It can be found that the three characteristic operating voltages  $V_{LQ}$ ,  $V_{\lambda_\pi}$ , and  $V_{\lambda_\sigma}$  have been shifted to  $\sim 157$  V,  $\sim 572$  V, and  $\sim 723$  V, respectively, which are still below the limit of 800 V for demonstrating our switchable laser system. Besides, the transmittances of all the corresponding high-Q and low-Q states (green dashed lines) remain comparable to those without considering the poling error (refer to Fig. 1(a)). To operate the EO APPLN device as an intracavity element in a dual-wavelength Nd:YLF laser, the two end (x) facets of the APPLN chip were polished and applied with optical coatings for highly transmitting ( $>99.5\%$ ) both 1047 and 1053-nm fundamental lines and for high transmittance ( $\sim 95\%$ ) of their second-harmonic/sum-frequency generation (SHG/SFG) waves, while the two y surfaces (1-mm spacing) were coated with nichrome as electrodes for the external electric-field ( $E_y$ ) application.

### 3. EO spectrum, polarization-state, and temporal-state (cw/pulsed) switchable laser source construction and characterization

Figure 4 shows the schematic experimental arrangement of the built EO switchable source in a cross-polarized, dual-wavelength Nd:YLF laser when operating with the implemented APPLN device. Nd:YLF features several unique material properties as introduced above, facilitating us to demonstrate in this study the manipulation of its two closely lying emission lines (1047 and 1053 nm) in the spectral, time, and polarization domains. On the other hand, however, Nd:YLF is characterized by a relatively low fracture limit ( $<40$  MPa) [32]. The thermal effects of a Nd:YLF crystal have been widely studied [33,34]. Several countermeasures to reduce the thermal loading of the laser crystal, as suggested in a study for power scaling of a Nd:YLF laser [34], have been employed in this work, including using a lower Nd doping concentration, a longer gain length, and an appropriately larger pump beam spot size. Accordingly, we adopted an a-cut, 0.8-at. % Nd:YLF crystal, having a length of 20 mm and a cross section of  $3 \times 3$  mm<sup>2</sup>, as the gain medium in this experiment, by referring to a Nd:YLF laser study with a moderate pump power level ( $\sim 10$  W) similar to ours [35]. To further facilitate the thermal treatment, the Nd:YLF crystal was wrapped with indium foil and fastened on a water-cooled copper mount, where the temperature is controlled by a thermoelectric cooler at 20 °C. The pump source for the laser system is a cw  $\sim 9$ -W (maximum), 806-nm fiber-coupled diode laser with a core diameter of 400  $\mu$ m. A focusing lens set was used to reimage the pump beam into the laser crystal. The 1047- and 1053-nm dual laser lines oscillate in an optical resonator configured by two dielectric mirrors, denoted as M1 and M2, where M1 is a 15-cm radius-of-curvature meniscus laser mirror having  $\sim 99.8\%$  reflectance at 1047 and 1053 nm and  $\sim 93\%$  transmittance at 806 nm, while M2 is a plane-plane laser mirror having  $\sim 95\%$  reflectance in the spectral range from 1040 to 1060 nm and  $>90\%$  transmittance in the frequency doubling range between 520 and 530 nm.

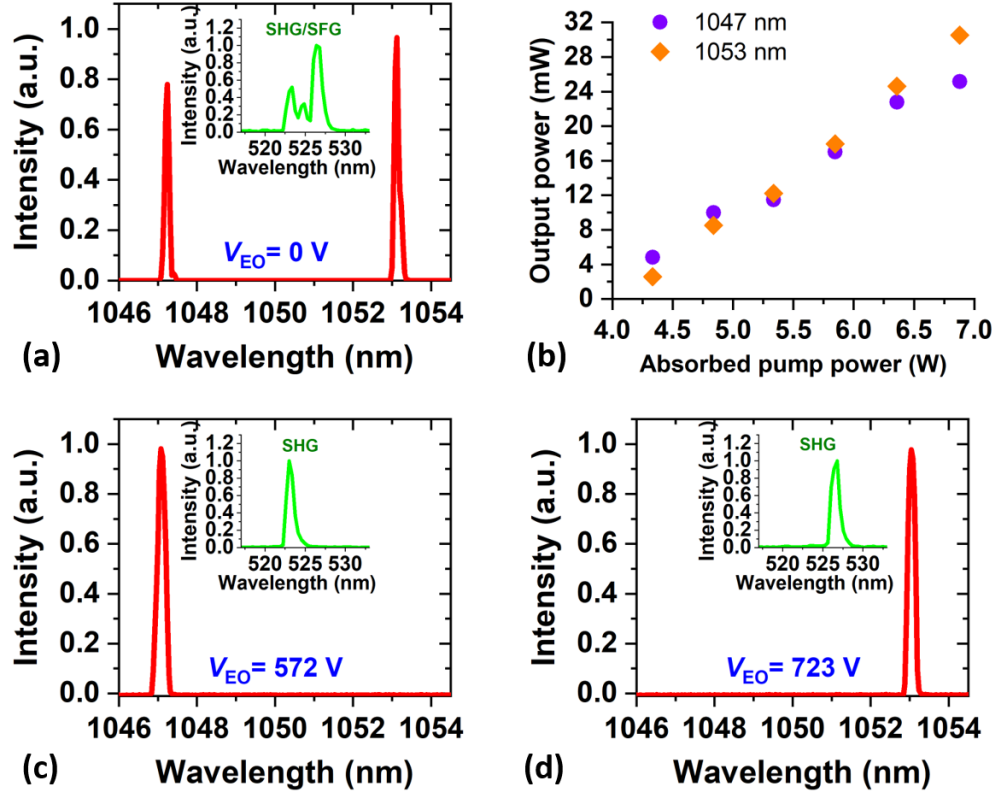


**Fig. 4.** Schematic arrangement of an EO spectrum, polarization-state, and temporal-state (cw/pulsed) switchable source built in a cross-polarized, dual 1047- and 1053-nm Nd:YLF laser in operation with an intracavity APPLN EO PMC chip.



Though the two Nd:YLF emission lines feature similar gain cross sections as introduced above, a careful alignment of the laser cavity setup, where we applied mainly via the fine adjustment of the translational and tilting degrees of freedom of M2, is required to achieve dual-wavelength oscillation. A removable polarizing beam splitter (PBS) is inserted right after the output coupler (M2) of the laser system, as schematically shown in Fig. 4, whenever separate characterization of the output performance of the two cross-polarized laser lines is desired. We first characterize the laser system in the cw operation mode. The  $M^2$  of the fundamental (1047 and 1053 nm) waves are similar, characterized to be  $\sim 1.6$  within the pump power range (up to 9 W, corresponding to an absorbed pump power of  $\sim 6.87$  W), using the beam profiling method [36] with the aid of an Infrared CCD camera. The beam spot sizes of the 1047 and 1053 nm waves in the laser crystal are estimated to be  $\sim 174$  and  $\sim 188 \mu\text{m}$ , simulated from the cavity geometry of the constructed dual-wavelength Nd:YLF laser system using the Gaussian beam propagation method and the characterized output beam quality. In the spectral measurements, a high-resolution ( $\sim 0.1$  nm) optical spectrum analyzer (Agilent 86142B) is used. However, its working wavelength range is limited to 600-1700 nm. We characterized the spectra of the SHG/SFG signals using a lower-resolution ( $\sim 2$  nm) miniature spectrometer (Ocean Optics USB2000 + XR1-ES) due to the availability in the lab. Figure 5(a) shows the measured output spectrum from the dual-wavelength Nd:YLF laser system at an absorbed pump power of  $\sim 6.87$  W, when the APPLN chip operates at 0 V at 30 °C. The threshold pump power for the dual-wavelength operation of the laser is observed to be  $\sim 4.33$  W, as shown in Fig. 5(b). Though the output power performance is not a major focus in this study, the power efficiency can be further improved by optimizing the output coupling ratios for both emission lines (the output coupler, M2, is used due to the availability in our lab) for reaching not only an optimal power coupling but also a balanced gain competition without relying much on an alignment of the laser cavity. A more careful laser cavity design to enhance the modal overlap between the pump and the fundamental beams inside the laser crystal can further improve the laser output efficiency. Additionally, the power scale-up of our system can be further enhanced through the use of several measures, such as low-quantum-defect pumping [37], a linear doping gradient design [38], and multi-segmented crystal structures [39]. Besides, the relatively small-aperture ( $0.5 \times 1 \text{ mm}^2$ ) and long (20 mm) APPLN chip will certainly limit the cavity tunability and energy throughput. Usually, a MgO-doped LN can be poled in a much thicker (a few mm) substrate and has a better power handling capability, which could be a potential solution for the issue. Nevertheless, we clearly demonstrated the first EO spectral switching in such a cross-polarized, closely-lying dual-wavelength laser system. Figures 5(c) and 5(d) show the measured output spectra of the Nd:YLF laser system when pumped at  $\sim 6.87$  W and the APPLN chip is activated with  $V_{\lambda\pi}=572$  V and  $V_{\lambda\sigma}=723$  V for selecting the  $\pi$ -polarized 1047 nm and  $\sigma$ -polarized 1053 nm lines, respectively, at 30 °C. We found the spectral switching can take effect in a range of the switching voltages for  $V_{\lambda\pi}$  between 549 and 581 V, and for  $V_{\lambda\sigma}$  between 715 and 732 V. Each EO switched spectrum in the built laser system can be obtained over a range of the applied voltage as long as the voltage-dependent laser gains for the dual laser lines, governed by the voltage tuning curve (Fig. 3(b)), meet both the laser oscillation and the respective spectral switching (Table 1) conditions. We also observed relatively weak, non-phase-matched second-harmonic and sum-frequency generations (SHG and SFG) from the system thanks to the high quadratic nonlinearity of LN, as shown in the insets of Figs. 5(a), 5(c), and 5(d) for the three EO spectral switching cases, respectively. Besides, an APPLN features relatively abundant spatial frequency components than a PPLN due to its aperiodic domain grating structure, potentially benefiting the enhancement of the compensation for the phase mismatching in these SHG/SFG processes. Appropriate bandpass filters have been used to conduct selective measurements of the fundamental lines and their up-conversion signals. By analyzing with the PBS, the 523.5-nm wave (SHG of the 1047-nm line) is found to be  $\pi$  polarized via the use of the nonlinear coefficient  $d_{33}$ , the 526.5-nm wave (SHG of the 1053-nm line) is  $\pi$

polarized with the access of the nonlinear coefficient  $d_{31}$ , and the 525 nm wave (SFG of the 1047 and 1053-nm lines) is  $\sigma$  polarized, produced based on the nonlinear coefficient  $d_{24}$ .

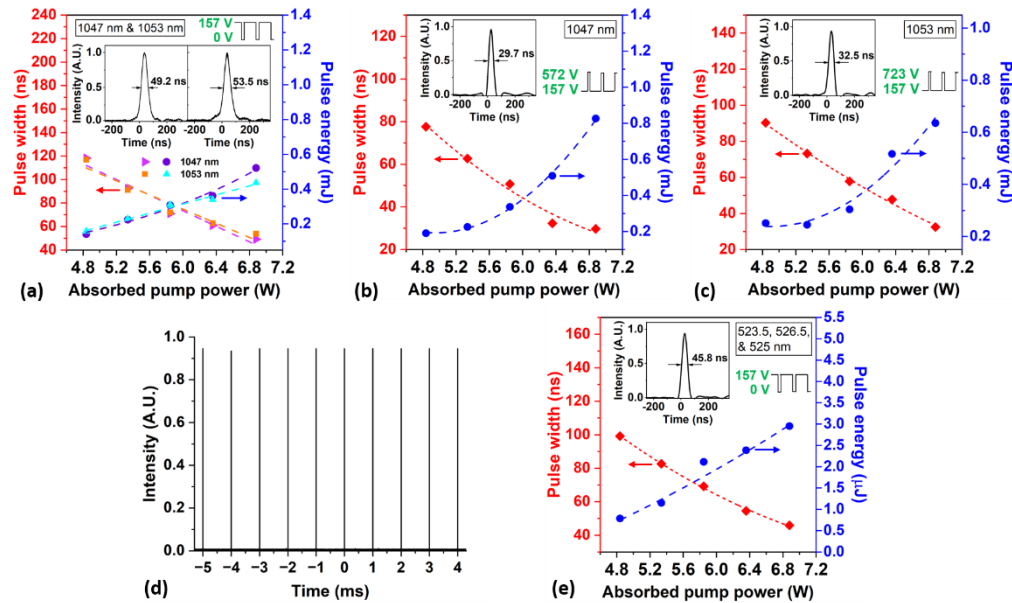


**Fig. 5.** Measured output spectra from the Nd:YLF laser system (see Fig. 4) in the cw mode at an absorbed pump power of  $\sim 6.87$  W, when the EO APPLN chip operates at  $V_{EO} =$  (a) 0 V, (c) 572 V, and (d) 723 V at 30 °C. The insets of the three EO switchable spectra show the corresponding SHG/SFG spectra, observed from the non-phase-matched nonlinear conversion of fundamental line(s) through the APPLN chip. (b) Measured output power of the system operating at dual wavelengths (i.e., when  $V_{EO} = 0$  V) as a function of the pump power.

In the same sense, an APPLN can be designed to provide multiple reciprocal vectors to quasi-phase-match multiple-frequency conversion processes simultaneously, in which, e.g., a multiwavelength source can be implemented [26,31]. It is then potentially possible to enhance the efficiency of these SHG/SFG waves by creating their associated QPM reciprocal vectors in the APPLN domain structure with our developed SA-AOS algorithm. However, it will inevitably compromise the optimization of  $G_{\lambda_{\pi}}$  and  $G_{\lambda_{\sigma}}$  in the APPLN with the algorithm, leading to the increase of the operating voltages.

To further demonstrate the laser pulse generation from the built system shown in Fig. 4, we adopted the Q-switching technique, where repetitively fast switching between a high-Q state and a low-Q state of the laser can be accomplished through the application of an appropriate electric pulse train to the intracavity APPLN. The electric pulses switch between two voltage levels,  $V_{HQ} = 0$ ,  $V_{\lambda_{\pi}}$ , or  $V_{\lambda_{\sigma}}$  (leading to the high-Q states of the laser for producing the three special spectra, respectively) and  $V_{LQ}$  (leading to a low-Q (gain-suppressed) state of the laser for both of the spectral lines), as concluded in Tab. 1 (see specifically the “Possible operating scheme” under the column “Q switching”). Figures 6(a), 6(b), and 6(c) show the measured output

pulse energy and pulse widths from the laser system as a function of the pump power when the APPLN is driven by a 1 kHz voltage pulse train of 300-ns duration with  $V_{HQ} = 0, 572$ , and  $723$  V, respectively, and  $V_{LQ} = 157$  V at  $30^\circ\text{C}$ . These three pulsed emissions feature not only different spectra but also different polarization states as discussed, radiating at dual (cross-polarized) 1047- and 1053-nm, single ( $\pi$ -polarized) 1047-nm, and single ( $\sigma$ -polarized) 1053-nm wavelengths, respectively. The insets of Figs. 6(a)–6(c) show the respective captured temporal pulses for the three EO switchable spectra from the laser system, indicating pulse widths on the order of a few tens of ns when pumped at  $\sim 6.87$  W. Accordingly, we estimate the output peak power for this pulsed EO switchable laser system can reach up to  $\sim 20$  kW under that pump level. It can be found that the detected pulses are based on a small ringing background. We suspect the origin of these (ns) ringing noises could be some external electromagnetic noises coupled from the homemade high-voltage (ns) pulse generator. Although an electromagnetic shielding has been implemented in our detection system (and actually effectively reduced these coupled noises), some residual background noise is still detectable. The signal-to-noise ratio (SNR) in these detection results is approximately 16 dB. We believe the SNR is actually much higher, as the optical pulse signals are so intense that we largely scatter the signals before sending them into the photodetector to avoid damage to the detection equipment. Besides using the electromagnetic shield, we are considering applying filters (such as adding filtering circuits) to both the pulser and photodetector electronics to further minimize unwanted noise signals.



**Fig. 6.** Measured output pulse energy and pulse widths from the Nd:YLF laser system (see Fig. 4) in the (Q-switched) pulsed mode as a function of the pump power when the EO APPLN is driven by a 1 kHz voltage pulse train of 300-ns duration with  $V_{HQ} =$  (a) 0, (b) 572, and (c) 723 V and  $V_{LQ} = 157$  V at  $30^\circ\text{C}$ . The insets show the detailed pulse information for the three EO Q-switched and spectrum switchable cases of the laser system. Note in (a), the results for the dual-wavelength Q-switched laser are presented separately at each individual wavelength. (d) Typical repetitive pulse train recorded from the laser output. (e) Measurement results for the SHG/SFG signals, produced by the Q-switched dual fundamental laser lines (Fig. 6(a)) through the APPLN chip.

The measurement results in Figs. 6(a)–6(c) show that the output performance of the Q-switched single 1047-nm line and single 1053-nm line obtained from the built laser system is similar, as

the cavity has been configured to approach a gain balance between the two lines to achieve the dual-wavelength oscillation condition initially (i.e., at 0 V), before any EO control of the laser gain. The still-existing discrepancy in pulse energy and width between the two pulsed emission lines can be mainly attributed to the technical difficulty in achieving an exact balance of gain between the two lines. On the other hand, the intervention of a gain/loss modulation effect from the intracavity EO APPLN in our system could be another factor causing the discrepancy in the output pulse performance. The transmittance of the APPLN is voltage dependent, as shown in Fig. 3, and could be slightly different when operating at the two characteristic voltages (i.e., at  $V_{\lambda\pi}=572$  V and  $V_{\lambda\sigma}=723$  V) for Q-switching the two cross-polarized lines, respectively.

When both lines are Q-switched, it can be expected that the available gain (initial inversion population density ( $N_i$ )) will be shared by the two transition lines, with a gain ratio depending on the level of the gain balance (or gain competition) between them. Since both the pulse energy and pulse width are sensitive functions of the initial gain with respect to the threshold gain ( $r \equiv N_i/N_{ic}$ , where  $N_{ic}$  is the critical inversion population density), the higher the  $r$ , the higher the pulse energy and the narrower the pulse width according to the laser Q-switching theory [8]. In this context, Q-switching on dual wavelengths will each produce a lower pulse energy and a broader pulsed width as  $N_i$  has to be shared, in contrast to the situation for Q-switching a single wavelength, where much less gain competition is expected. Our measurement results, shown in Fig. 6, agree with the above analysis.

A thorough theoretical and experimental study on a dual-wavelength EO Q-switched laser has been reported elsewhere [40]. In that study, based on timing control of two EO Q-switches working respectively on dual-wavelength 1064- and 1319-nm lines in a Nd:YAG laser, the simulated pulse energy and pulse width of the test line (1319 nm) exhibit enhanced and narrowed, respectively, with increasing the delay time between the two Q-switches, which supports a consistent result with our experimental observation and the analysis made above.

Figure 6(d) shows a typical repetitive pulse train recorded from this APPLN-based EO switchable Nd:YLF laser, exhibiting a peak-to-peak fluctuation of  $\sim 2\%$ , which is relatively small than that reported in [41]. This could be attributable to a moderate pump power level that we have used in the system. It is interesting to find that in our system, an inferior peak-to-peak intensity fluctuation (around 5-10%) can be observed when the EO switching is made from one spectrum to another spectrum among the three featured output spectra (i.e., the dual-wavelength and the two single-wavelength emissions), especially when switching is made from the single-wavelength operation to the dual-wavelength one. A slight adjustment of the cavity alignment (within a range that remains supporting a balanced gain condition) is required to achieve a more stable operation with a  $\sim 2\%$  peak-to-peak fluctuation after every switching (note that fast switching is still demonstrated without an extra cavity tuning, but with a relatively low intensity stability). We suspect that an optimized setup (cavity configuration) for dual-wavelength oscillation may differ slightly from that for single-wavelength operation at 1047 or 1053 nm. This interesting phenomenon also evidences the gain competition between the  $\pi$ -polarized 1047 nm laser and the  $\sigma$ -polarized 1053 nm laser, and its sensitivity to the cavity conditions.

Limited by the electronic specifications of our homemade voltage pulser employed in this work, the fastest pulse repetition rate is 1 kHz. Since the Q-switching mechanism developed in this study is based on the EO effect, we expect the repetition rate of the built pulsed laser system to have considerable room for increase, if desired. We again observed the emission of non-phase-matched SHG/SFG accompanying those fundamental pulses. These SHG/SFG were obtained with similar output pulse energy (on the order of a few  $\mu\text{J}$ ) or peak power (on the order of  $\sim 100$  W). We represently present the measurement results for the pulsed 523.5, 526.5, and 525-nm signals simultaneously converted from the Q-switched dual 1047- and 1053-nm fundamental waves (Fig. 6(a)), as shown in Fig. 6(e). The results demonstrate the achievement of not only the EO Q-switching for pulse generation, but also the EO spectral switching, and

simultaneously, polarization-state switching among pulses with dual 1047- and 1053-nm, single 1047-nm, and single 1053-nm wavelengths. This novel system shows its great potential for multiplexability in time, wavelength, and polarization.

The technique developed in this study can be directly applied to an integrated photonic platform, such as bulk or thin-film LN waveguides, to effectively reduce the footprint and switching voltages of the system.

#### 4. Conclusion

We have demonstrated, to the best of our knowledge, the first compact solid-state laser system switchable in temporal, spectral, and polarization degrees of freedom. The system is realized in a cross-polarized, dual 1047- and 1053-nm Nd:YLF laser in operation with an intracavity EO APPLN chip. The APPLN, designed using the AOS technique with the aid of the SA optimization method, works as a dual-wavelength EO PMC to modulate the polarization-dependent laser gain of the Nd:YLF, in which three different spectra of different polarization states, dual (cross-polarized) 1047- and 1053-nm lines, single ( $\pi$ -polarized) 1047-nm line, and single ( $\sigma$ -polarized) 1053-nm line can be acquired from the novel laser system simply by switching the APPLN operating voltage to  $V_0 = 0$  V,  $V_{\lambda_\pi} = 572$  V, and  $V_{\lambda_\sigma} = 723$  V, respectively. A special system state can be obtained when operating at  $V_{LQ} = 157$  V, where both laser lines are totally suppressed. Take advantage of this unique operating scheme, we further achieve the spectral switchable Nd:YLF laser system in (ns) pulsed operation mode via the Q-switching technique when the APPLN fast repetitively switches between a high-Q-state voltage (at  $V_{HQ} = 0, 572$ , or  $723$  V) and a common low-Q-state voltage (at  $V_{LQ} = 157$  V). The output peak power level can reach  $\sim 20$  kW in such a pulsed EO switchable laser system with a diode pump power of  $\sim 6.87$  W. Thanks to the aperiodic domain structure and high nonlinearity of the APPLN device, the demonstration of the temporal, spectral, and polarization switching behavior in the fundamental lines of the Nd:YLF laser system is also observed in their accompanying non-phase-matched SHG/SFG.

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**Disclosures.** The authors declare no conflicts of interest.

**Data availability.** Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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